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An Interpretable Predictive Model for Liquid Holdup of Hilly Terrain Oil-Gas Pipelines Based on Machine Learning and Shapley Additive Explanations

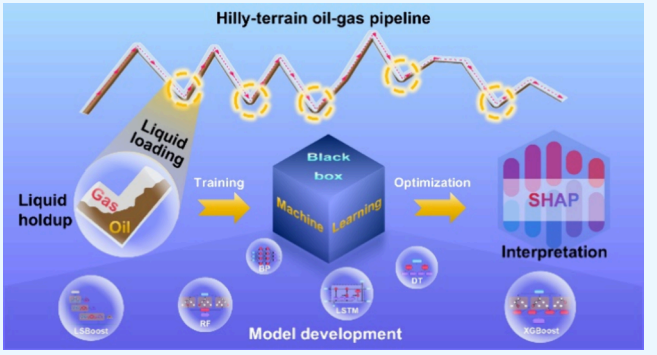
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ABSTRACT: During oil and gas field development, hilly terrain pipelines for mixed oil and gas transportation often face challenges from liquid loading, which significantly impacts flow assurance. However, the intricate nonlinear relationship between liquid loading and pipeline system operations complicates the solution of liquid holdup in practical pipelines by using traditional mechanistic models. Research combining machine learning (ML) for liquid loading prediction is limited, and most studies focus solely on validating the accuracy and efficiency of ML models without addressing their interpretability. To address this issue, an interpretable predictive model integrating ML and Shapley additive explanations (SHAP) is proposed. Operating conditions for an actual hilly terrain oil-gas pipeline were simulated using OLGA software to generate a comprehensive data set. Six ML models were compared using four evaluation metrics to assess their predictive performance for liquid holdup. eXtreme Gradient Boosting (XGBoost) demonstrated the best performance, with an R^2 of 0.986, MAE of 0.012, MAPE of 0.030, and RMSE of 0.016 on the test set. The XGBoost model was then visualized using SHAP to provide interpretability. Specifically, global explanations were given to ascertain the contribution of individual features to the average liquid holdup in low-lying areas of the pipeline. Local explanations were also conducted for individual pipeline operation data samples, visualizing the influence of each feature on the average liquid holdup within any specific sample alongside their cumulative effects. The interactions among the features were analyzed to derive trends in the interaction effects of different pipelines and operating conditions on the average liquid holdup over different ranges.



1. INTRODUCTION

The expanding development of oil and gas fields has driven the rapid advancement of multiphase pipeline infrastructure, ensuring efficient transportation of crude oil and natural gas from extraction sites to processing plants, storage depots, and consumption hubs worldwide. The integration of oil and gas transportation within a single pipeline system not only enhances operational efficiency but also minimizes energy resource wastage, thereby highlighting its critical importance in the energy sector and optimal resource utilization. However, oil and gas mixed transportation pipelines face severe liquid loading issues in hilly terrain environments.[1,2] During pipeline transportation, factors such as variations in oil and gas flow rates and pipeline slopes can lead to the accumulation of liquid phases within the pipeline, reducing its flow capacity and transportation efficiency, and even posing potential safety hazards.[3,4] These hazards include the risk of hydrate formation causing blockages, potential acidic components causing pipeline corrosion, and resultant leakage risks.[5,6] Especially in complex terrain conditions, such as mountainous regions with significant undulations, the liquid loading problem during mixed oil and gas transportation is even more severe. Therefore, accurate prediction of liquid loading plays a vital role in the design and optimization of hilly terrain oil-gas pipelines and their operating conditions.

The formation of liquid loading in oil-gas pipelines is strongly correlated to the dynamics of two-phase flow. Traditionally, two primary approaches have been used to predict liquid holdup in two-phase pipelines: empirical correlations and mechanistic modeling. Early research employed empirical correlation methods to study the liquid holdup patterns of horizontal and inclined pipelines.[7−10] In recognition of the limitations in accuracy and application scope

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of purely empirical calculations, researchers have advanced a variety of models built upon the mechanisms of two-phase flow, such as the Taitel and Dukler model,[11] the Shoham and Taitel model,[12] the Chen model,[13] and Fan's two-fluid model.[14] Solving these models requires consideration of different specific working environments, which can be computationally complex.

Advancements in numerical computation technology have enabled significant progress in multiphase flow simulations, offering effective tools for liquid loading prediction.[15] Currently, some scholars used numerical simulation methods to study multiphase flow in pipelines, including using OLGA software to replicate multiphase flow experiments in pipelines and compare experimental data on liquid holdup[16−18] and pressure changes,[19−21] simulating the patterns of liquid loading in pipelines under different loading conditions,[22,23] and simulating hydrate formation in long-distance pipelines under complex working conditions.[24] In most practices, including the aforementioned studies, the accuracy of the simulation results obtained using the OLGA software has been well-validated. It can simulate multiple operating conditions and facilitate the study of complex patterns of liquid loading influenced by multiple factors. However, some multiphase flow simulation software, including OLGA, typically requires parameter settings for specific operating conditions during each simulation, which means that a significant amount of modeling and computation time is required when discussing a large number of complex and variable operating scenarios. Therefore, there is a need to actively explore new methods that can predict liquid loading in mixed transportation pipelines more efficiently and accurately.

With the rise of big data science and artificial intelligence, ML offers new methodologies and insights for pipeline engineering, finding extensive applications in monitoring and predicting pipeline leaks,[25−28] corrosion,[29−31] faults,[32] vibrations,[33,34] strengths,[35,36] and other aspects. The problem of liquid loading in mixed flow pipelines involves complex, nonlinear, and multifactor comprehensive effects. Recently, some academics have explored multiphase flow in mixed flow pipelines based on various ML algorithms. Hong et al.[22] applied Random Forest (RF) to train data derived from OLGA simulation, aiming to predict where liquid loading occurs in simplified wet gas pipelines. Liu et al.[37] developed an RF model for forecasting liquid holdup, employing 2,755 data sets related to two-phase flow experiments conducted by four researchers. Xiao et al.[38] constructed a backpropagation neural network (BPNN) model based on 300 sets of horizontal pipeline experimental data to accurately predict the liquid holdup of horizontal pipelines. Rastogi and Fan[39] presented a method that synthesizes ML and mechanical models to estimate both pressure gradients and holdup. Zhang et al.[40] put forward a physically guided neural network technique aimed at forecasting pressure drop and holdup in two-phase pipelines. Aforementioned studies have demonstrated that they outperform traditional empirical and analytical methods in terms of accuracy and efficiency when predicting liquid holdup in mixed flow pipelines. Nonetheless, these ML models operate as black boxes, leading to uncertainty regarding the influence of input features on output predictions and interactions among features. Therefore, making accurate and efficient ML models interpretable is highly meaningful for understanding the mechanisms of data interaction, which can provide direction for the design and determination of operating conditions of oil and gas pipelines, thereby achieving the goal of increasing production while reducing transportation risks. Recently, a small amount of research in the field of pipeline engineering has explored novel methods for using Shapley additive explanations (SHAP) to explain ML models, including five aspects: burst pressure,[41] pressure drop,[42] maximum corrosion depth,[43,44] defect weld identification,[45] and failure pressure.[46] Research in each of these areas has demonstrated SHAP's ability to make ML models transparent, providing a clear basis for pipeline design, management, and risk assessment. However, it can be found from the above research that there are fewer studies on combining ML to achieve liquid holding prediction, and the training of models in them is mainly based on experimental data with specific limitations. Moreover, these ML models focus on verifying only the accuracy and efficiency of their predictions for liquid holdup, and the lack of interpretability hinders their further application.

To address the aforementioned issues and overcome limitations, this study focuses on developing an interpretable predictive model for liquid holdup of hilly terrain oil-gas pipelines through the integration of ML and SHAP. First, the oil and gas mixed transportation process of an actual hilly terrain pipeline under various combinations of operating conditions was simulated using OLGA software, and a large amount of operational data was obtained and processed to establish a database. Then, the predictive performance of six ML models, including BPNN, RF, XGBoost, decision tree (DT), long short-term memory (LSTM), and least squares boosting (LSBoost), for liquid holdup was compared based on four evaluation metrics. Finally, the XGBoost model that achieved the optimal predictive performance was visualized through SHAP, utilizing its interpretable mechanism. The impact of various input features on the average liquid holdup of pipelines was explained at the global level, and the contribution of each feature to the average liquid holdup of pipeline operation data samples was visualized at the local level. The interaction between features and the average liquid holdup of the pipelines was analyzed.

The subsequent sections are outlined as follows. Section 2 presents the methodology for predicting liquid holdup of hilly terrain oil-gas pipelines based on interpretable ML, including data collection, ML algorithms, performance metrics, and SHAP. Section 3 conducts a comparison of six ML models regarding their predictive performance for liquid holdup of hilly terrain oil-gas pipelines and selects the best model for in-depth discussion on the interpretability analysis of its predictions. Lastly, Section 4 outlines the conclusions derived from this research.

2. METHODOLOGY

2.1. Data Collection. Predicting the liquid loading in actual hilly terrain oil-gas pipelines is a complex challenge. For a determined actual pipeline route, liquid loading will be affected by factors such as pipeline diameter (D), gas-oil ratio (GOR), outlet pressure (P_{out}), inlet temperature (T_{in}), etc.[47−49] Due to the influence of terrain, gravity exerts negative work on the liquid phase in the upward sloping section of undulating mixed flow pipelines during operation, resulting in a decrease in liquid flow velocity and significant liquid accumulation. As the accumulation persists over time and the transportation distance increases, the liquid level in the low-lying sections rises, adversely affecting the transportation efficiency and creating a range of safety risks for pipeline

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operations. Therefore, this study suggests employing the average liquid holdup at multiple low-lying locations along a defined pipeline route to characterize the extent of liquid loading across various operating conditions.

In this study, the OLGA multiphase flow simulation software is utilized to construct a liquid loading prediction model tailored to the actual route and operating conditions of a hilly terrain oil-gas pipeline in the Changqing Oilfield, which has been proven reliable through extensive validation and application by numerous scholars.[50−52] In OLGA, the elevations and lengths are configured according to the actual pipeline routing, as depicted in Figure 1. The pipeline

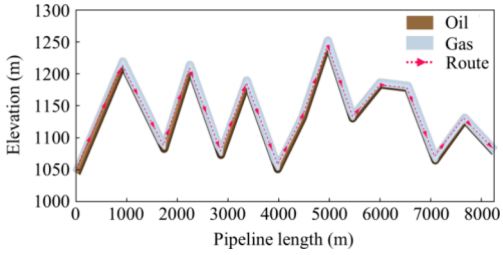


Figure 1. Routing of an actual hilly terrain oil-gas pipeline.

diameter, gas/oil ratio, outlet pressure, and inlet temperature are set within practical ranges, with specific parameters outlined in Table 1. Additionally, gas has a specific gravity of

Table 1. Parameter Settings for Numerical Simulation Model of Liquid Loading

symbol	parameter	range
D	pipeline diameter (mm)	60, 90, 120, 150, 180
GOR	gas-oil ratio	40, 60, 80, 100, 120
P_{out}	outlet pressure (MPa)	0.20, 0.26, 0.32, 0.38, 0.44, 0.50
T_{in}	inlet temperature (°C)	40, 50, 60, 70, 80

0.65, oil has a specific gravity of 0.9, and the liquid flow rate is 160 Sm^3/d. A comprehensive set of 750 operating scenarios is established based on the parameter ranges. Drawing upon the simulation results from OLGA, for each scenario, liquid holdup at multiple low-lying points along the pipeline is exported for each scenario, followed by the calculation of their average values. The organized simulation data are presented in Figure 2. Preliminary observations from Figure 2 reveal that the average liquid holdup varies significantly with different gas/oil ratios, followed by pipeline diameters. In contrast, there is minimal variation in average liquid holdup data under different outlet pressures and inlet temperatures. Specifically, when the GOR is 40, the average liquid holdup tends to cluster more around (0.6, 0.7], whereas at a GOR of 120, it leans toward (0.2, 0.3]. In terms of pipeline diameter, the trend shows that as the diameter increases, the lowest average liquid holdup initially decreases and then increases, with the largest diameter yielding an average liquid holdup predominantly within (0.5, 0.6]. Across different outlet pressures and inlet temperatures, the average liquid holdup predominantly falls within (0.3, 0.4]. Among all samples, the maximum average liquid holdup recorded is 0.75, the minimum is 0.23, and the overall average stands at 0.44.

2.2. ML Models. *2.2.1. BPNN.* BPNN, as a classical method of multilayer feed-forward neural network training, was first proposed by Rumelhart and McClelland et al.[53] and has been widely used. The basic principle of BPNN is shown in Figure 3a, which receives the data through the input layer, processes the complex information through one or more hidden layers, and finally produces the prediction result via the output layer. In a BPNN, data is forward propagated through the connections between layers, while the BP algorithm adjusts the weights based on the errors calculated, reducing the discrepancies between the expected and observed outputs.

2.2.2. DT. DT is a powerful ML technique employing a tree-like structure to recursively partition a data set in order to predict the values of continuous type variables, as illustrated in Figure 3b.[54] The algorithm constructs internal nodes by selecting the best features, with each node representing a single test of an attribute and branches directing to different subsets based on the test results. As the data are continuously subdivided, each leaf node eventually represents a prediction region and outputs the constant values within that region. This construction process aims to maximize the prediction accuracy of each segmented subset until a stopping condition is met.

2.2.3. LSTM. LSTM, introduced by Hochreiter and Schmidhuber,[55] is a specialized variant of a recurrent neural network (RNN) that focuses on overcoming the gradient vanishing or exploding challenges encountered by traditional RNNs when processing long sequence data. The LSTM model consists of multiple memory cells, each of which is controlled by a forget gate, input gate, and output gate, as illustrated in Figure 3c. The forget gate evaluates which information from the previous cell state should be discarded, while the input gate determines the new information to be added, and the output gate specifies the components of the cell state that will be outputted.

In Figure 3c, C, h, and X represent the cell state, output, and input, respectively; The subscripts t-1 and t indicate the preceding and current time points; σ signifies the sigmoid function; $\otimes$ denotes the process of information filtering, often implemented through the sigmoid function to determine which parts of the input should be let through (or filtered out); $\oplus$ symbolizes the process of information fusion, where filtered and transformed inputs are combined with the previous cell state to produce a new cell state; tanh is the activation function with a range of $[-1,1]$.

2.2.4. RF. RF is an ensemble learning algorithm introduced by Breiman,[56] incorporating multiple DTs as its core components. It performs regression and classification tasks by constructing numerous DTs and aggregating their predictions. The algorithm operates by first randomly drawing multiple subsets from the original data set using the bootstrap sampling method. Each of these subsets is then used to independently train a decision tree. These DTs, acting as individual base learners, made predictions independently. Their outputs are subsequently averaged (for regression) to generate the final predictive value, thereby forming a strong learner model known as a Random Forest, whose structure is shown in Figure 3d.

2.2.5. LSBoost. LSBoost is an ML algorithm that can be used to predict continuous numerical data, based on the least-squares method as the loss criterion.[57] This algorithm is particularly effective for regression problems. It iteratively constructs a strong regression model consisting of multiple weak regression models, each of which is a decision tree. In each iteration step, the model's predictive capability is progressively enhanced by fitting the residuals generated by the current iteration. Optimization is achieved by minimizing

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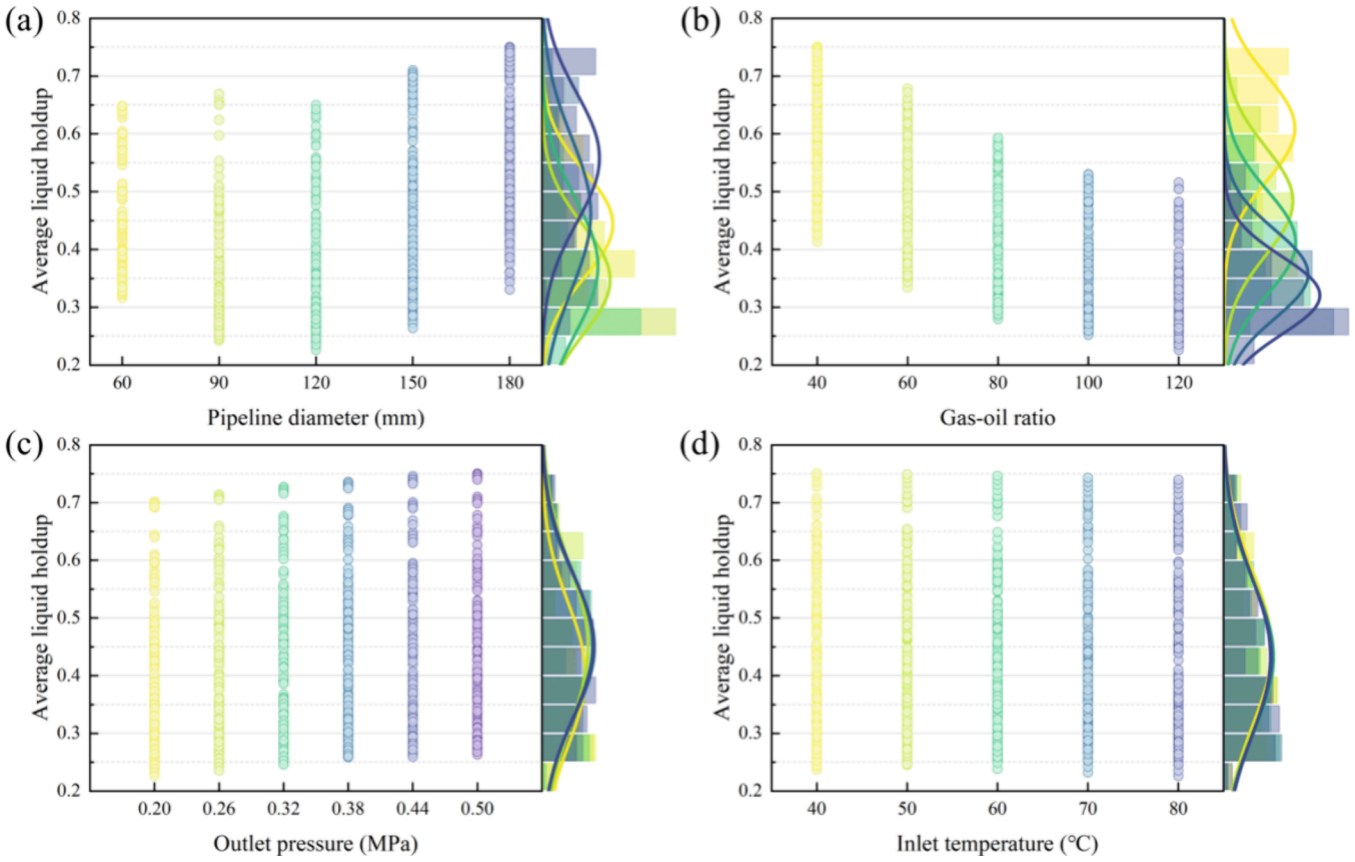


Figure 2. Distribution of average liquid holdup under different (a) pipeline diameters, (b) gas-oil ratios, (c) outlet pressures, and (d) inlet temperatures in the data set.

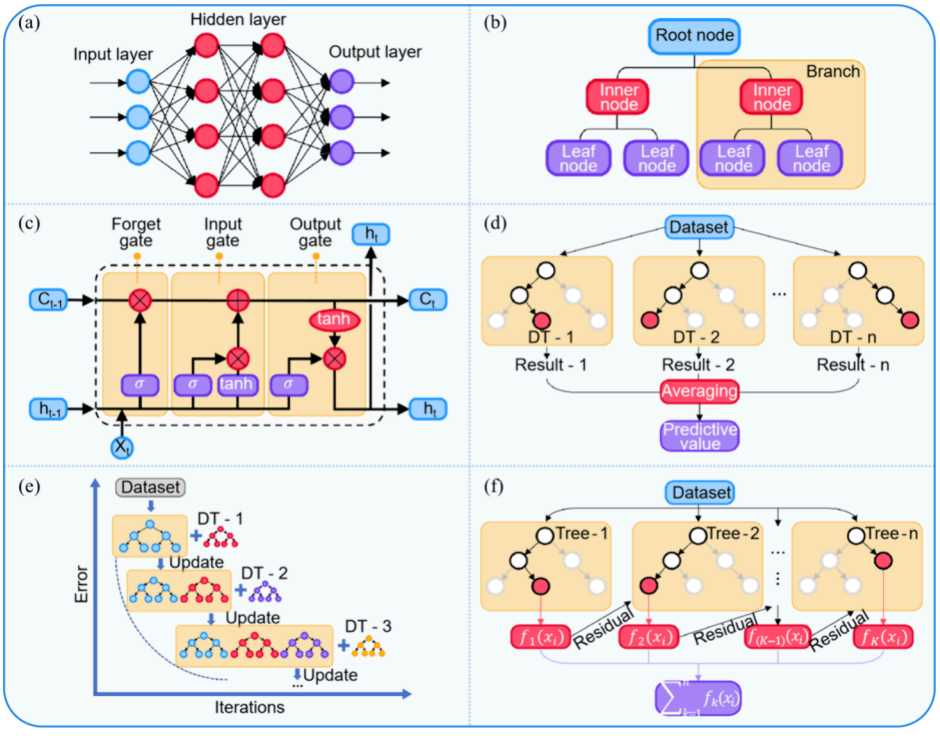


Figure 3. General structures of (a) BPNN, (b) DT, (c) LSTM, (d) RF, (e) LSBoost, and (f) XGBoost.

the mean square error (MSE), thus narrowing the gap between predicted and actual values. LSBoost's structure is presented in Figure 3e.

2.2.6. XGBoost. XGBoost, developed by Chen et al.,[58] is an enhanced algorithm for gradient boosting decision trees, which plays an important role in ensemble learning methods. Its

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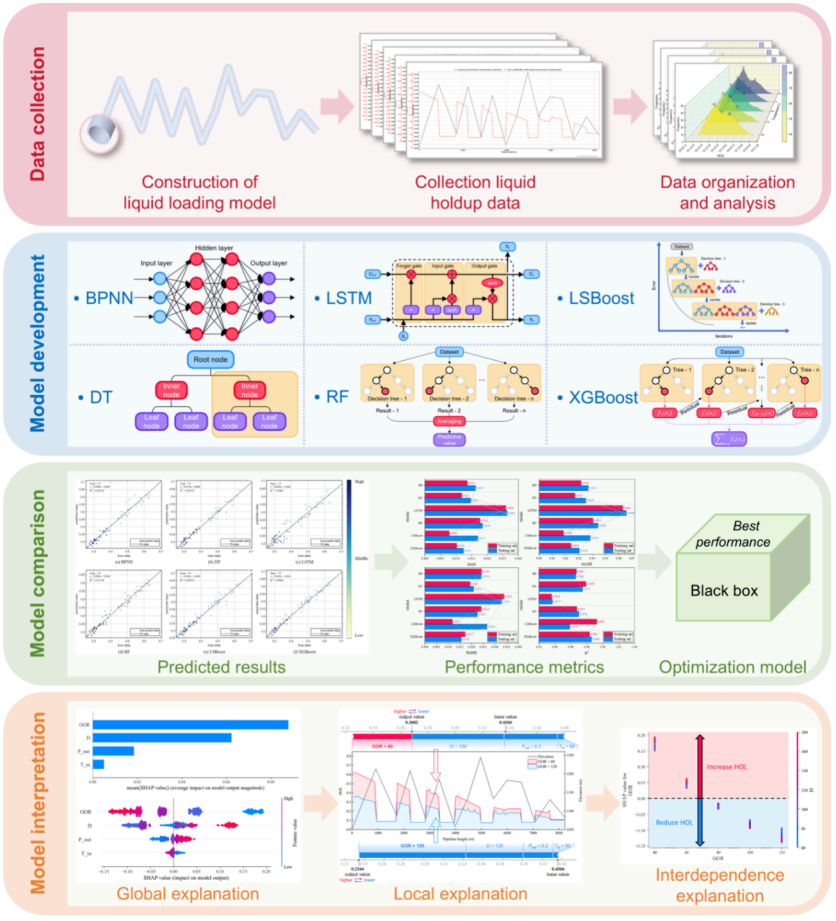


Figure 4. Framework diagram illustrating the four steps of constructing interpretable ML: data collection, model development, comparison, and interpretation.

structure is shown in Figure 3f. XGBoost uses a combination of multiple weak learners (DTs) to construct a strong prediction model. Each subsequent DT aims to reduce the residual of the previous tree, improving the prediction results through gradual iteration and ultimately constructing a robust prediction model. In addition, Additionally, XGBoost strikes a balance between model complexity and generalization by incorporating regularization terms into the loss function. By using this approach, the potential for overfitting is significantly diminished, leading to more accurate predictions.

2.3. Performance Metrics of ML Models. For evaluating the predictive performance of ML models regarding liquid holdup of hilly terrain oil-gas pipelines, four classical metrics were computed, namely, MSE, root MSE (RMSE), mean absolute error (MAE), and goodness of fit (R^2). The mathematical expressions of the above performance metrics were given in eqs 1−4, respectively:

$$\mathrm{MSE} = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2 \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2} \tag{2}$$

$$\mathrm{MAE} = \frac{1}{m}\sum_{i=1}^{m}|y_i - \hat{y}_i| \tag{3}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{m}(y_i - \bar{y}_i)^2} \tag{4}$$

where y_i and $\hat{y}_i$ denote the actual and the ML-predicted average liquid holdup of the (i)-th group of samples; $\bar{y}_i$ refers to the mean value of the actual average liquid holdup of the samples; and m stands for the total count of samples.

The metrics MSE, RMSE, and MAE all provide insights into the degree of dispersion between the actual and the ML-predicted average liquid holdups of a model. A smaller value for any of these metrics (closer to 0) signifies a lower deviation between the two values, thereby indicating a better predictive performance. While R^2 measures the degree of fit between the two values, with a normal range of R^2 being [0, 1], values approaching 1 indicate a superior fit and better predictive performance.

2.4. SHAP. ML is widely utilized in scientific research due to its robust predictive capabilities and proficiency in handling complex nonlinear data. However, ML models are often black boxes lacking interpretability. To address this limitation, Lundberg and Lee, inspired by cooperative game theory, introduced SHAP.[59] SHAP functions as a posthoc approach for explaining model predictions, whose core idea lies in calculating the expected marginal contribution (Shapley value) of feature variables to the predictions of ML. It then offers explanations for ML from both a global and local standpoint.[59,60] At the global level, SHAP determines the primary

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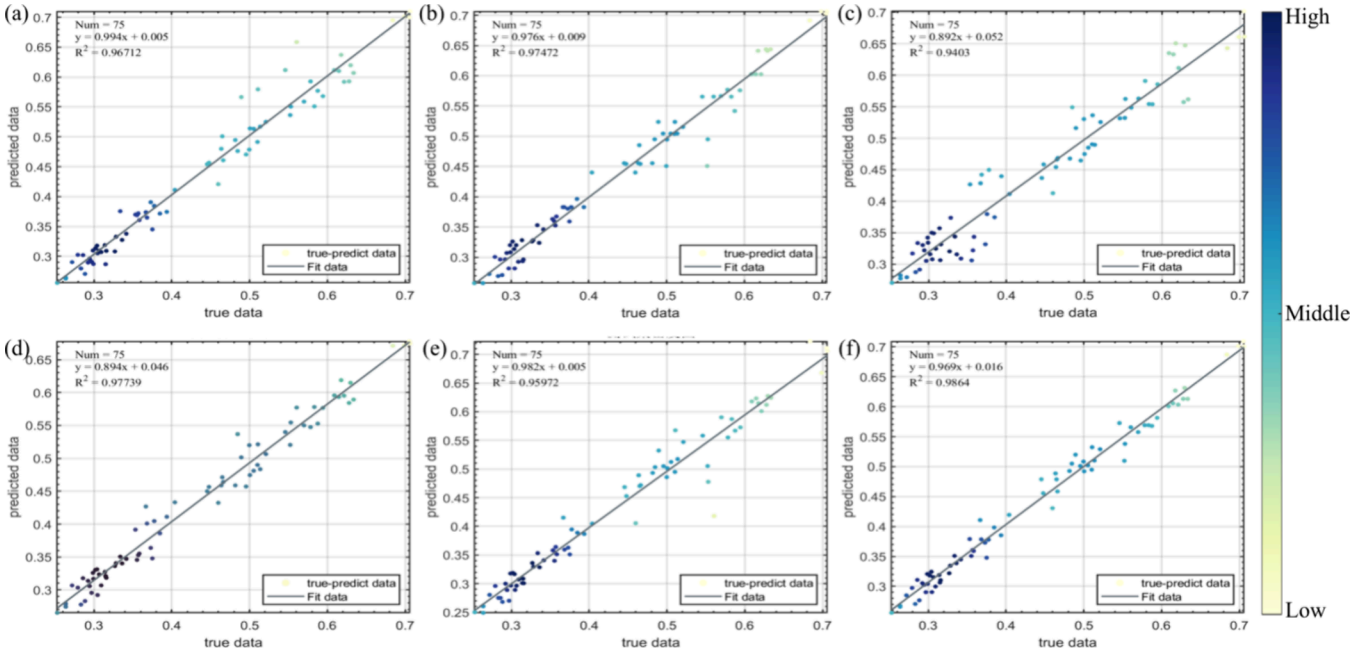


Figure 5. Scatter density plot of predicted and true values for (a) BPNN, (b) DT, (c) LSTM, (d)RF, (e) LSBoost, and (f) XGBoost.

influence of individual features on the ML's predictions, allowing for a deeper understanding of the relative importance of various factors. Locally, SHAP explores specific predictions, revealing how individual features affect certain cases. SHAP has proven to be a valuable tool for interpreting ML models across various domains.[61−63] In this study, we introduce SHAP to gain deeper insights into how ML models predict the liquid holdup of hilly terrain oil-gas pipelines based on different features and to rank and explain the importance of these features. SHAP interprets the Shapley value as an additive feature attribution approach, where the ML's prediction is explained as the sum of attribution values for each input feature, represented as a linear function of binary variables, as follows[59]:

$$g(z') = \phi_0 + \sum_{j=1}^{n} \phi_j z'_j \tag{5}$$

where g represents the interpretation model; $z' \in \{0,1\}^M$ denotes the coalition vector, indicating the presence or absence of corresponding features; n stands for the total input feature count; ϕ_0 refers to the average output of the interpretation model; ϕ_j indicates the SHAP value assigned to feature j.

The SHAP value for feature j is calculated using the following equation:[59]

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(n - |S| - 1)!}{n!} [f(S \cup \{j\}) - f(S)] \tag{6}$$

where S represents a subset of all features except for feature j; $|S|$ represents the size of the set; N stands for the full set of features; $f(S)$ denotes the prediction of the ML model's prediction for a given subset S, and $f(S \cup \{j\}) - f(S)$ is the difference caused by feature j.

Additionally, SHAP allows for the investigation of interaction effects between pairs of features by computing SHAP interaction values.[60] These SHAP interaction values are derived by combining SHAP values with Shapley interaction indices from game theory[64]:

$$\phi_{j_1,j_2} = \sum_{S \subseteq N \setminus \{j_1,j_2\}} \frac{|S|!(n - |S| - 2)!}{2(n-1)!} \nabla_{j_1,j_2}(S)$$

$$\text{when } j_1 \neq j_2,\ \nabla_{j_1,j_2}(S) = f_x(S \cup \{j_1, j_2\}) - f_x(S \cup \{j_1\}) - f_x(S \cup \{j_2\}) + f_x(S) \tag{7}$$

Addressing the issue of the time-consuming exact calculation of SHAP values, Lundberg et al.[64] provided an ensemble algorithm named TreeSHAP, which significantly reduces the complexity of computing SHAP values exponentially, thereby accelerating the computation process. In this study, the SHAP library in Python is used to determine SHAP values.

2.5. Framework for Interpretable ML. The framework outlined in this study for the interpretable ML model is depicted in Figure 4, structured around four key steps: (1) data collection, as described in Section 2.1, involves constructing a liquid loading model using OLGA to obtain liquid holdup data, which is then organized and analyzed before being divided into training, validation, and testing sets; (2) six ML models using different algorithms with their own advantages are established; (3) comparison of the prediction results and performance metrics for these ML models is conducted, which enables the selection of the model that demonstrates the highest performance as the optimal one; and (4) global, local, and interaction interpretations of the optimal model are conducted using SHAP to explain the mechanism of action of each feature.

3. RESULTS AND DISCUSSION

3.1. Performance Comparison among Various ML Models. The 750 sets of data simulated using the OLGA software were divided into training, validation, and testing sets in an 8:1:1 ratio, utilizing pipeline diameter, gas/oil ratio, outlet pressure, and inlet temperature as inputs, with average liquid holdup as the output. Predictive models were trained

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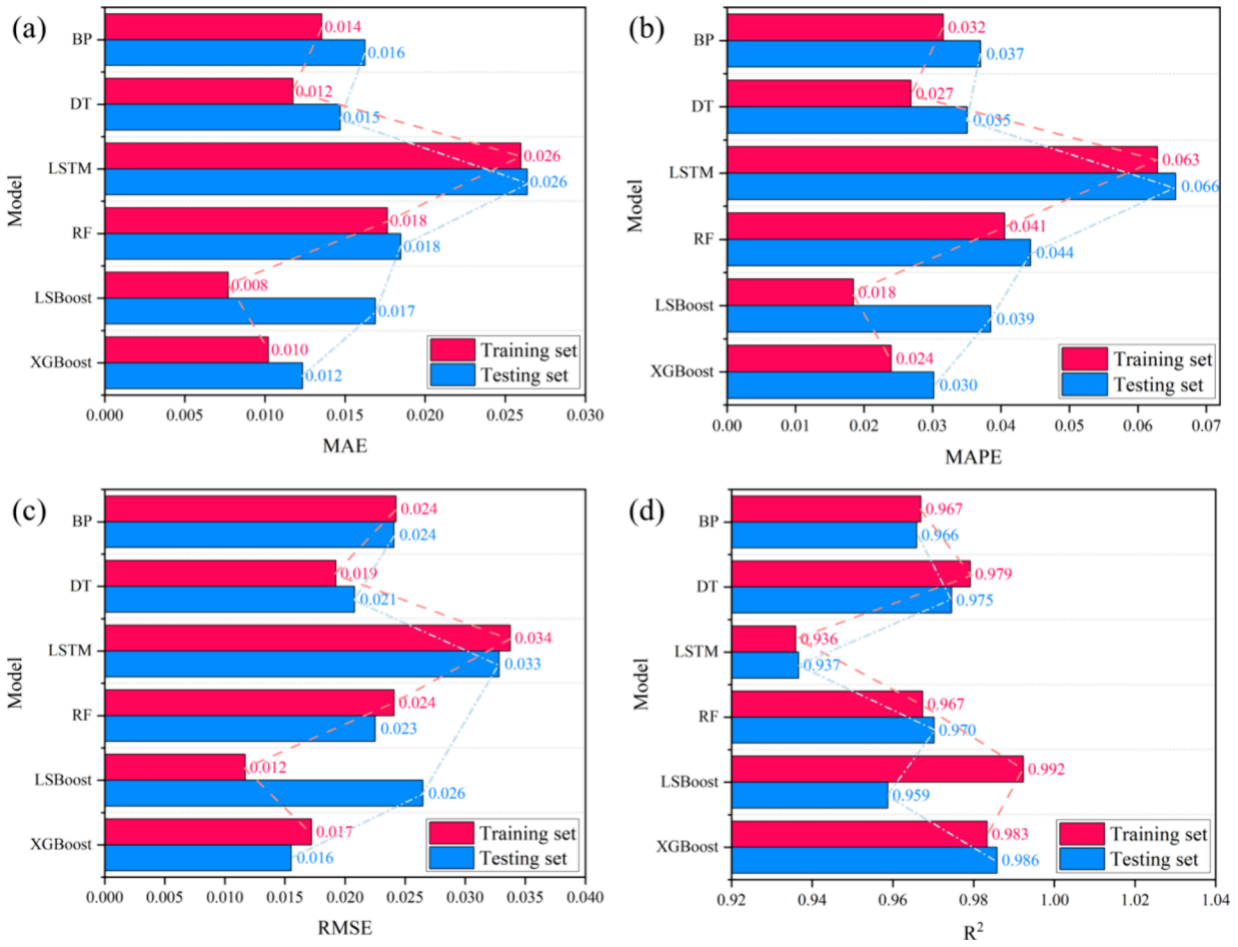


Figure 6. Comparison of predictive performance of six ML methods through metrics: (a) MAE, (b) MAPE, (c) RMSE, and (d) R^2.

through BPNN, DT, LSTM, RF, LSBoost, and XGBoost, and predictions were made on the testing set. Figure 5 presents a comparison of the predicted and actual average liquid holdup of the six ML models. It can be observed that the testing set includes samples with average liquid holdup between 0.25 and 0.7, mainly concentrated in the interval of 0.3−0.4. Overall, the fitting effect of each model is excellent, indicating that these models demonstrate great accuracy in predicting the liquid holdup of hilly terrain oil-gas pipelines.

Figure 6 displays the results of the evaluation for the predictive performance of six ML models, assessed through four metrics. For the training set, the R^2 values for all six models exceeded 0.9, signifying excellent predictive performance. Among them, LSBoost and XGBoost achieved the best fits with R^2 values exceeding 0.98, while LSTM had the worst fit with an R^2 value of 0.936. In terms of MAE, MAPE, and RMSE, LSBoost exhibited the lowest evaluation values, followed by XGBoost, while LSTM had the highest. Analysis of these four metrics for the training set suggests that LSBoost and XGBoost models have the best predictive performance, while LSTM has the worst. For the testing set, among the six ML models, XGBoost had the lowest MAE, MAPE, and RMSE values and the highest R^2 value of 0.986. However, LSBoost exhibited a marked decline in predictive performance when applied to the testing set, with its R^2 dropping from 0.992 on the training set to 0.959, which is lower than those of XGBoost, DT, RF, and BPNN. In terms of MAE, MAPE, and RMSE, LSBoost ranked among the top three highest values, which indicate poorer predictive performance. Among the models tested, XGBoost, DT, and RF performed the best, while LSTM still performed worst. A comprehensive analysis of the predictive performance of the six models on both the training and testing sets reveals that all models have good predictive capabilities. Among them, XGBoost performed best, while LSTM performed worst. In this research, XGBoost emerged as the optimal model for predicting liquid holdup of hilly terrain oil-gas pipelines. The superior performance of XGBoost compared to the other models can be attributed to several factors. First, XGBoost is known for its ability to handle complex, nonlinear relationships in the data due to its boosting mechanism, which sequentially corrects the errors made by previous models. Additionally, it effectively reduces overfitting through regularization techniques and has robust handling of missing data and feature importance. These advantages allow XGBoost to consistently deliver higher predictive accuracy across multiple evaluation metrics. Further interpretation of its predictive results was conducted to gain a deeper understanding of its predictive capabilities.

3.2. Model Interpretation. *3.2.1. Global Explanation.* SHAP was utilized to evaluate the importance of four influencing factors, resulting in a ranking of their impact, as illustrated in Figure 7. The findings revealed that gas-oil ratio and pipeline diameter had the greatest impact, followed by outlet pressure, while inlet temperature had the least influence. The SHAP global interpretation plot in Figure 8 uses color to indicate the feature values for real samples. Red corresponds to

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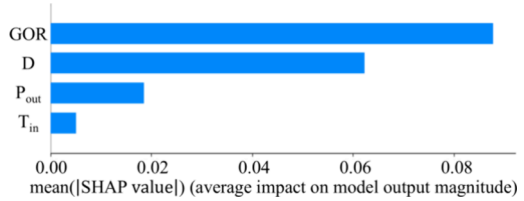


Figure 7. Feature importance bar plot shows the average contribution of each feature on the predicted average liquid holdup in XGBoost, determined by the mean of absolute SHAP values.

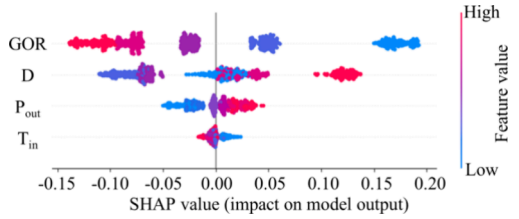


Figure 8. SHAP bee swarm plot, where each point corresponds to a sample and the color scale reflects the magnitude of the feature value.

higher values, blue to lower values, and purple to values near the average. The SHAP value is shown along the horizontal axis, where values above zero point to a stronger positive effect on the prediction, and values below zero suggest a stronger negative influence.

Figure 8 shows a color transition of the GOR from red to purple to blue moving left to right, suggesting that as the GOR decreases, its positive effect on the average liquid holdup increases. In other words, a smaller GOR leads to a higher average liquid holdup and a more severe problem of liquid loading in the pipeline. The relationship between the pipeline diameter and average liquid holdup does not show a clear positive or negative correlation. However, excessively large pipeline diameters have a significant positive effect on average liquid holdup, causing it to increase. Conversely, pipeline diameters that are close to the average size in the data set have a negative effect, reducing average liquid holdup. When the pipeline diameter is small, the SHAP values are near zero, indicating that smaller pipeline diameters result in average liquid holdup rates that are close to the average under the operating conditions of this data set. For outlet pressure, an increase in outlet pressure leads to an increase in the average liquid holdup. As for the inlet temperature, a decrease in inlet temperature promotes an increase in average liquid holdup, while higher temperatures inhibit this increase. However, overall, the impact of temperature changes on the average liquid holdup is relatively small.

3.2.2. Local Explanation. SHAP values can explain individual prediction outcomes, offering insights into the interpretation of single data points. Two samples are extracted and illustrated in the upper and lower bands of Figure 9, respectively, with the starting point representing the base value, which serves as the model's baseline. Band colors represent feature contributions, and the length of each band corresponds to the strength of that feature's influence on the final prediction. A red band reflects a positive feature contribution that increases the prediction value, while a blue band indicates a negative contribution, reducing it. In this study, the

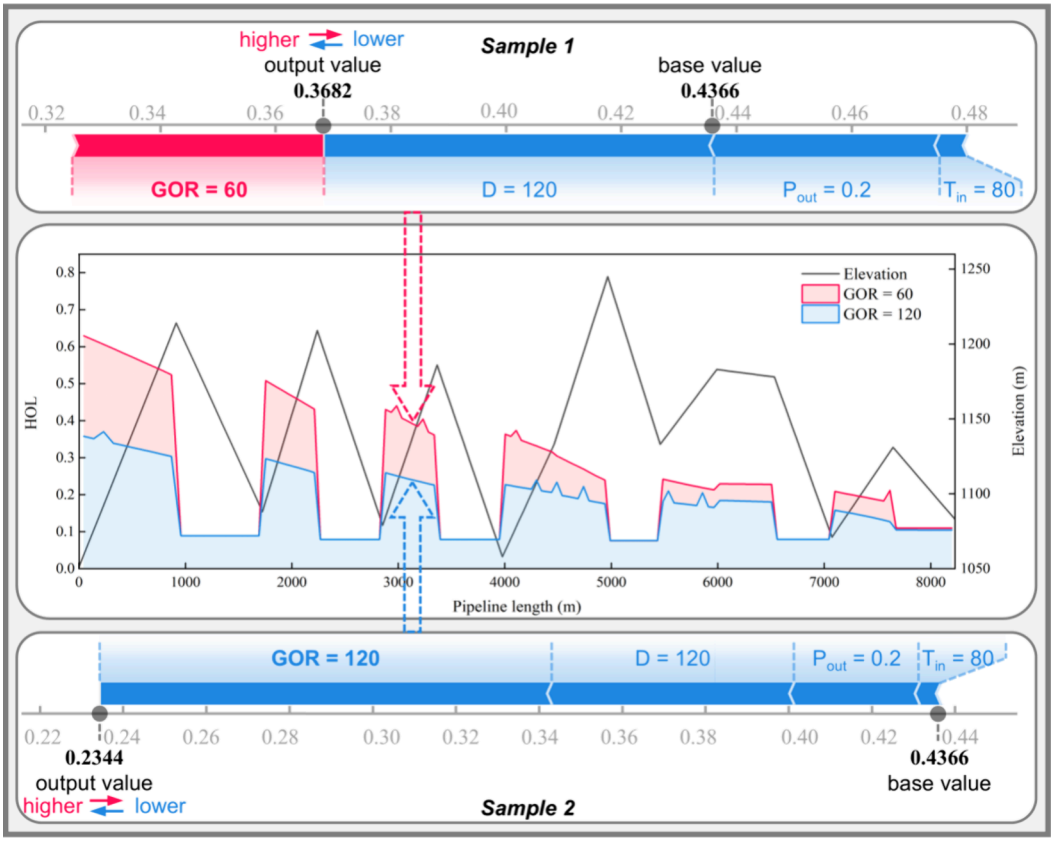


Figure 9. Force plot used for local interpretation: taking two samples with only different gas-oil ratios as examples and displaying the distribution of liquid holdup along the hilly terrain oil-gas pipeline.

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prediction value corresponds to the average liquid holdup forecasted by the optimal model, XGBoost, which is calculated by adding the baseline value to the contributions from all of the features. The prediction values for the upper and lower samples in Figure 9 are 0.23 and 0.37, respectively. The two samples differ only in the GOR, with both having a pipe diameter of 120 mm, an outlet pressure of 0.2 MPa, and an inlet temperature of 80 °C. By comparison, when the GOR is 120, all features contribute negatively to the average liquid holdup, with the GOR having the largest contribution. Considering the combined effects of all features, the final average liquid holdup is 0.2344, indicating that a higher GOR plays a primary role in reducing the liquid holdup. When the GOR changes to 60, the pipeline diameter contributes the most to reducing the liquid holdup, while the GOR in this sample increases the average liquid holdup. Considering the combined effects of all features, the final average liquid holdup is 0.3682. A more detailed analysis can be conducted by retrieving the along-pipeline liquid holdup for these two samples from the original simulation result database, as shown in the middle part of Figure 9. The results indicate that the highest liquid holdup occurs in low-lying areas, and in uphill sections, it is greater at a GOR of 60 compared to 120. In the first three low-lying areas, the liquid holdup decreases by approximately 40% when the GOR increases from 60 to 120. This method provides a clear depiction of how each feature contributes to the predictions for individual samples. Using SHAP to uncover local characteristics further validates the previously mentioned conclusion regarding the impact of the GOR's effect on average liquid holdup.

3.2.3. Interdependence Explanation. Utilizing SHAP to illuminate the interaction between the two most influential features, namely, the GOR and pipeline diameter, as depicted in Figure 10, each point represents an actual sample. On the *x*-

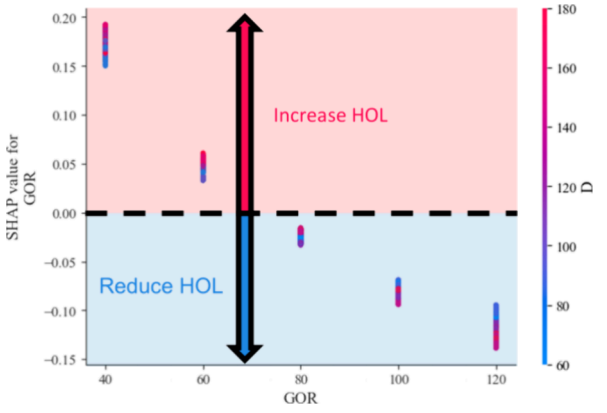


Figure 10. Feature dependency graph: taking the two most important features (gas oil ratio and pipeline diameter) as an example.

axis, the magnitude of the GOR is illustrated, and the color of the points reflects the size of the pipeline diameter, where redder shades denote larger diameters and bluer shades signify smaller ones. The vertical axis depicts the SHAP value under the interaction. When the GOR is 40 or 60, regardless of the pipeline diameter ranging from 60 to 180 mm, their interaction contributes to an increase in the average liquid holdup, with larger diameters enhancing this promotional effect. Conversely, when the GOR is 80, 100, or 120, their interaction negatively impacts the average liquid holdup across the same range of pipe diameters. Additionally, when GOR is 80, an increase in pipeline diameter may facilitate an increase in liquid holdup, whereas when GOR is 120, a decrease in pipeline diameter promotes an increase in liquid holdup. By interpreting the interactions between features using SHAP, one can clearly observe the interactive effects of different features within various ranges on the predicted value, providing a visual method for understanding the roles of each feature. This offers a reference basis for understanding field operating conditions and pipeline diameter design.

4. CONCLUSIONS

This study proposes an interpretable predictive model based on ML and SHAP, capable of reliably predicting and analyzing the liquid holdup in hilly terrain oil-gas pipelines. Operational data from actual pipelines were simulated using OLGA to establish a data set, which was then analyzed for average liquid holdup under various operating conditions. Six ML models were constructed to train and test the data set, and the optimal model was selected based on a comparison of four evaluation metrics. Furthermore, by integration with SHAP, the predictions of the selected ML model were provided with global, local, and interactive interpretations. The following conclusions were drawn:

1. Operational data for actual hilly terrain oil-gas pipelines under different gas-oil ratios, pipeline diameters, outlet pressures, and inlet temperatures were calculated using the OLGA simulator. The severity of liquid loading in the pipelines was reflected by the average liquid holdup at each low-lying point. Among all 750 samples, the maximum, minimum, and mean values of the average liquid holdup were 0.75, 0.23, and 0.44, respectively. The sample data distribution showcased the average liquid holdup under various features, allowing for preliminary but not precise inferences about the impact of each feature.
2. Six ML models were explained in principle, applied, and evaluated. In terms of prediction performance, LSBoost and XGBoost were the top performers on the training set, while XGBoost and DT excelled on the test set. After comprehensively comparing the six ML models, XGBoost and LSTM demonstrated the best and worst performance, respectively. Specifically, XGBoost achieved an R^2 of 0.986, MAE of 0.012, MAPE of 0.030, and RMSE of 0.016 on the test set.
3. XGBoost was recognized as the optimal choice after a comprehensive assessment of various models and integrated with SHAP to construct an interpretable ML model. Global interpretation through SHAP demonstrated how four features affect the average liquid holdup. Local interpretation of individual pipeline operational data samples using SHAP visualized the contribution of each feature to the average liquid holdup in any given sample, providing a reference direction for optimizing pipeline operation for specific samples. Through analysis of feature interactions, trends in the interactive effects of different pipeline and operating conditions on average liquid holdup within various ranges were identified.
4. The proposed XGBoost-SHAP model accurately predicts the liquid holdup in hilly terrain oil-gas pipelines and visualizes the ML black box through an interpretable

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mechanism. This method exhibits excellent generalizability. While this study focused on predicting and interpreting pipeline liquid loading, the predicted and interpreted content could also be extended to other areas of pipeline engineering, such as pipeline corrosion, leakage, wax deposition, and so forth.

5. The sample size for simulation data is relatively limited and solely derived from OLGA software calculations, lacking on-site operational data. Future research will aim to include field data samples to enhance the predictive precision and reliability of the proposed model. The study focuses on a specific pipeline system and does not consider the influence of more complex terrain variations. Future research will incorporate factors such as pipeline inclination and elevation differences to develop a more comprehensive liquid holdup prediction model under coupled conditions.
6. The current study evaluates the performance of a subset of machine learning models, but the comparison with baseline methods remains incomplete. Future work should extend the analysis to include a wider range of algorithms, such as CatBoost, SVR, and LightGBM, to comprehensively assess their potential in improving liquid holdup predictions. Additionally, exploring the applicability of emerging machine learning techniques may further enhance the robustness and versatility of the prediction framework.

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Notes

The authors declare no competing financial interest.

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